Evaluating Style Transfer for Text
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As an effective medium that portrays the equivalence of sentence structures via word choices, style transfer is perhaps among the fewer, acknowledged NLP techniques that still lack an adequate evaluation system to its credit. NLP literature introduces the complexities involved in this process, and the reader is apprised of the challenges and redressals it entails. As with any evaluation process, the suitability of metrics is first called into question and the report judiciously addresses it by introducing the novelties of relative measures (instead of absolute ones) and a custom-style lexicon. With the improvised metrics and newly chosen definitions of Style Transfer Intensity, Content Preservation and Naturalness, the automated evaluation goes on to exhibit heightened correlation with human judgment. And while this may signal a successful calibration of the metrics, to the extent of concluding this investigation, the audience is nevertheless drawn to examine the finer details of this innovation. Relative measures are inherently diagnosed with the notion of 'conditionality', i.e. the style transfer intensity or naturalness measured on any text are conditional on the input source. In the event that the source sentences themselves are ridden with flaws, the resulting relative measures stand invalidated. Thus, although the synchronization with human evaluation measures may persuade us to believe that the model has served its purpose, in reality, the transferability of this measure into an absolute context is still questionable.

The decision to use a statistical pattern-based classifier such as logistic regression in various parts of the study, could proceed with caution. While promising to be the all-encompassing standard reference for style words, a style lexicon still demands a robust underlying mechanism to help generate the possible style drivers. Using a logistic regression model to this extent might limit the potential of such a paradigm, considering how inappropriate data necessarily engenders improper models. For instance, the presence of contradictory training samples such as sarcasm, ironies, or sentences that have likely content words 'masquerading' as style words, could mislead the model into choosing the wrong style words. To the degree that content preservation depends on the masking of near-to-accurate style words in samples, this could potentially produce false evaluations. On similar grounds, supervised learning methods have also been noted for their reliance on labeled data for adequate performance. With the generation of a sufficient number of style words contingent on the availability of surplus labeled samples, it may be reckoned that the lexicon developed could underestimate the actual style words present across the corpora. An incomprehensive dictionary may then become the initiator of erroneous content preservation metrics described in the study. The use of logistic regression to ascertain the naturalness of text rekindles a similar deliberation. As its learning process constrains the classifier's results, the effects of sample sentences exhibiting counter-intuitive results also apply here (sarcasm, though unnatural by semantics, is more natural compared to a style-reversed sarcasm). Scrutiny over the data samples used in this research could help avoid possible deceptive metric evaluations encountered in this study.